



Lecture 6: Power Factor Improvement

EEE 321: Power System I

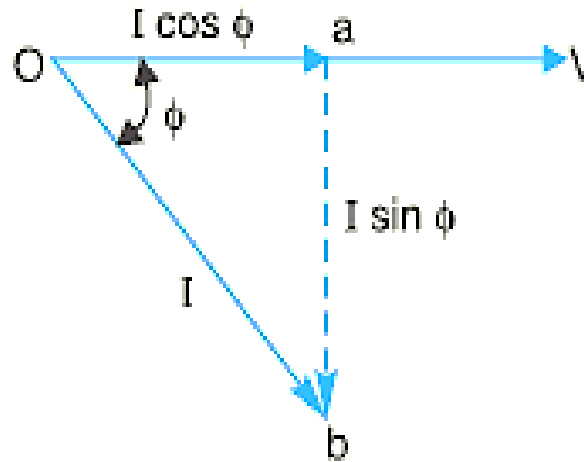
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Presentation Outline

- ❑ Power Factor and Power triangle
- ❑ Causes of low power factor
- ❑ Disadvantages of low power factor
- ❑ Importance of power factor Improvement
- ❑ Methods for Improving Power factor
- ❑ Examples and problems

Power Factor

In an a.c. circuit, there is generally a phase difference ϕ between voltage and current. The term $\cos\phi$ is called the power factor of the circuit.



It is a usual practice to attach the word 'lagging' or 'leading' with the numerical value of power factor to signify whether the current lags or leads the voltage. Thus if the circuit has a p.f. of 0.5 and the current lags the voltage, we generally write p.f. as 0.5 lagging.

If the circuit is inductive, the current lags behind the voltage and the power factor is referred to as lagging. However, in a capacitive circuit, current leads the voltage and power factor is said to be leading.

Consider an inductive circuit taking a lagging current I from supply voltage V ; the angle of lag being ϕ . The phasor diagram of the circuit is shown in Fig.

Power Factor

The circuit current I can be resolved into two perpendicular components, namely ;

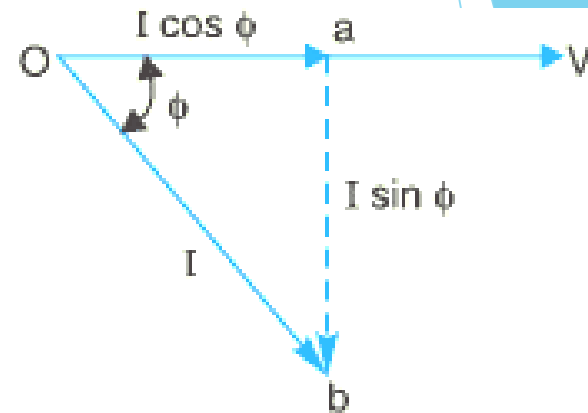
(a) $I \cos \phi$ is in phase with V

(b) $I \sin \phi$ is 90° out of phase with V

The component $I \cos \phi$ is known as active or wattful

component whereas component $I \sin \phi$ is called the reactive

or wattless component.



The reactive component is a measure of the power factor. If the reactive component is small, the phase angle ϕ is small and hence power factor $\cos \phi$ will be high. Therefore, a circuit having small reactive current (i.e., $I \sin \phi$) will have high power factor and vice-versa. It may be noted that value of power factor can never be more than unity.

Power Factor

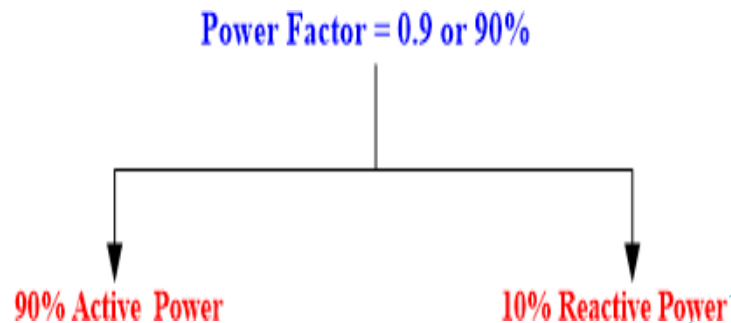
Power factor is a measure of how efficiently electrical power is consumed.

Power factor tells us how much value for money we are getting for the power that we consume.

Example: Suppose power factor of your house is 0.9 lagging (or) 90% lagging. What does this suggest?

It suggests that out of 100% power that is being supplied to you by the supplier, only 90% of the power is useful power (or) active power that is being utilized to run all the appliances of your house. The remaining 10% power is reactive power that is useless for you but still you have to pay for it.

It is not the concern of the supplier that how much power is useful for you and how much is wasted. You have to pay for the entire 100% power.



Power Triangle

An electrical power is composed of different parts:

kW – Working Power (also called Actual Power or Active Power or Real Power):- It is the power that actually powers the equipment and performs useful work.

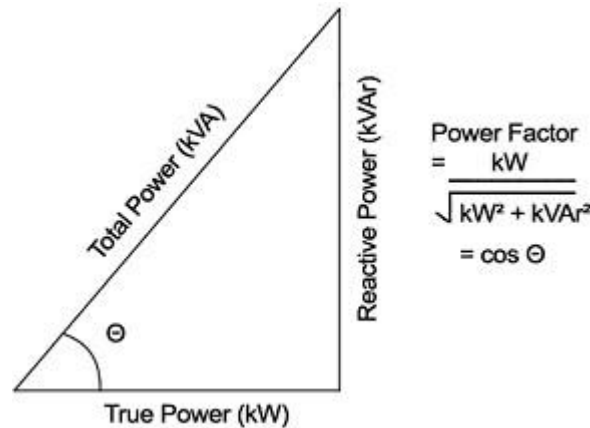
kVAR – Reactive Power:- It is the power that magnetic equipment (transformer, induction motor, and relay) needs to produce the magnetizing flux.

kVA – Apparent Power:- It is the “vectorial summation” of KVAR and KW.

Power Triangle illustrates this relationship between KW, KVA, KVAR, and

Power Factor: $PF = KW/KVA = \cos\phi$

$KVAR/KVA = \sin\phi$



Causes of Low Power Factor

Low power factor is undesirable from economic point of view. Normally, the power factor of the whole load on the supply system is lower than 0.8. Since power factor is defined as the ratio of KW to KVA, we see that low power factor results when KW is small in relation to KVA and causes a large KVAR in a system.

The following are the causes of low power factor:

- (i) Inductive loads draw reactive power (KVAR) from the electrical system. Most of the a.c. motors are of induction type (1 ϕ and 3 ϕ induction motors) which have low lagging power factor. These motors work at a power factor which is extremely small on light load (0.2 to 0.3) and rises to 0.8 or 0.9 at full load.
- (ii) Arc lamps, electric discharge lamps and industrial heating furnaces operate at low lagging power factor. These inductive loads constitute a major portion of the power consumed in industrial complexes.
- (iii) The load on the power system is varying; being high during morning and evening and low at other times. During low load period, supply voltage is increased which increases the magnetisation current. This results in the decreased power factor.

Disadvantages of Low Power Factor

The power factor plays an importance role in a.c. circuits since power consumed depends upon this factor.

$$P = V_L I_L \cos \phi \quad (\text{For single phase supply})$$

$$\therefore I_L = \frac{P}{V_L \cos \phi} \quad \dots(i)$$

$$P = \sqrt{3} V_L I_L \cos \phi \quad (\text{For 3 phase supply})$$

$$\therefore I_L = \frac{P}{\sqrt{3} V_L \cos \phi} \quad \dots(ii)$$

It is clear from above that for fixed power and voltage, the load current is inversely proportional to the power factor. Lower the power factor, higher is the load current and vice-versa.

A power factor less than unity results in the following disadvantages :

(i) Large kVA rating of equipment. The electrical machinery (e.g., alternators, transformers, switchgear) is always rated in kVA.

$$\text{kVA} = \frac{\text{kW}}{\cos \phi}$$

It is clear that kVA rating of the equipment is inversely proportional to power factor. The smaller the power factor, the larger is the kVA rating. Therefore, at low power factor, the kVA rating of the equipment has to be made more, making the equipment larger and expensive.

Disadvantages of Low Power Factor

(ii) Greater conductor size. To transmit or distribute a fixed amount of power at constant voltage, the conductor will have to carry more current at low power factor. This necessitates large conductor size.

For example, take the case of a single phase a.c. motor having an input of 10 kW on full load, the terminal voltage being 250 V. At unity p.f., the input full load current would be $10,000/250 = 40$ A. At 0.8 p.f.; the kVA input would be $10/0.8 = 12.5$ and the current input $12,500/250 = 50$ A. If the motor is worked at a low power factor of 0.8, the cross-sectional area of the supply cables and motor conductors would have to be based upon a current of 50 A instead of 40 A which would be required at unity power factor.

(iii) Large copper losses. The large current at low power factor causes more I^2R losses in all the elements of the supply system. This results in poor efficiency.

(iv) Poor voltage regulation. The large current at low lagging power factor causes greater voltage drops in alternators, transformers, transmission lines and distributors. This results in the decreased voltage available at the supply end, thus impairing the performance of utilization devices. In order to keep the receiving end voltage within permissible limits, extra equipment (*i.e.*, voltage regulators) is required.

(v) Reduced handling capacity of system. The lagging power factor reduces the handling capacity of all the elements of the system. It is because the reactive component of current prevents the full utilisation of installed capacity.

Importance of Power Factor Improvement

The improvement of power factor is very important for both consumers and generating stations as discussed below:

(i) For consumers: A consumer has to pay electricity charges for his maximum demand in kVA plus the units consumed. If the consumer improves the power factor, then there is a reduction in his maximum kVA demand and consequently there will be annual saving due to maximum demand charges. Although power factor improvement involves extra annual expenditure on account of p.f. correction equipment, yet improvement of p.f. to a *proper value* results in the net annual saving for the consumer.

(ii) For generating stations: A generating station is as much concerned with power factor improvement as the consumer. The generators in a power station are rated in kVA but the useful output depends upon kW output. As station output is $\text{kW} = \text{kVA} \times \cos \phi$, therefore, number of units supplied by it depends upon the power factor. The greater the power factor of the generating station, the higher is the kWh it delivers to the system. This leads to the conclusion that improved power factor increases the earning capacity of the power station.

Methods for Improving Power Factor

Consider an inductive load taking a lagging current I at a power factor $\cos \phi_1$. In order to improve the power factor of this circuit, the remedy is to connect such an equipment in parallel with the load which takes a leading reactive component and partly cancels the lagging reactive component of the load. Fig. 6.6 (i) shows a capacitor connected across the load. The capacitor takes a current I_C which leads the supply voltage V by 90° . The current I_C partly cancels the lagging reactive component of the load current as shown in the phasor diagram in Fig. 6.6 (ii). The resultant circuit current becomes I' and its angle of lag is ϕ_2 . It is clear that ϕ_2 is less than ϕ_1 so that new p.f. $\cos \phi_2$ is more than the previous p.f. $\cos \phi_1$.

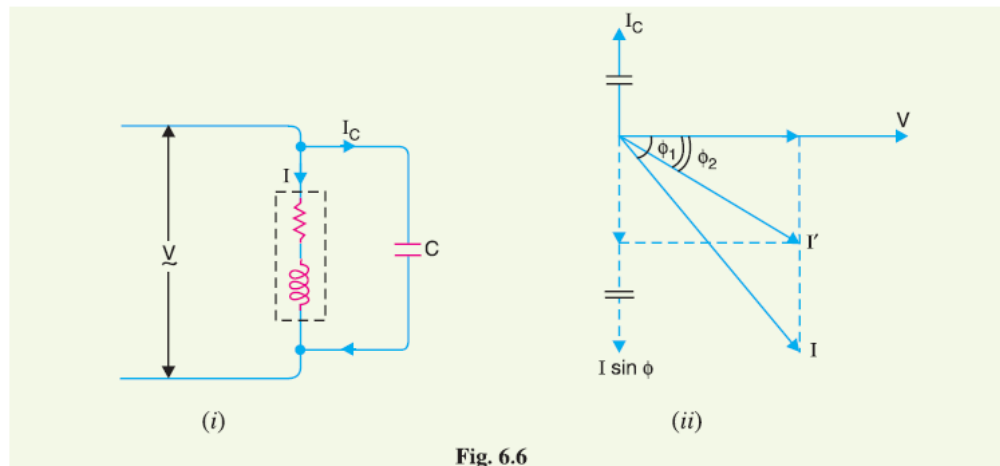


Fig. 6.6

From the phasor diagram, it is clear that after p.f. correction, the lagging reactive component of the load is reduced to $I' \sin \phi_2$.

Obviously,
$$I' \sin \phi_2 = I \sin \phi_1 - I_C$$

or
$$I_C = I \sin \phi_1 - I' \sin \phi_2$$

$\therefore$ Capacitance of capacitor to improve p.f. from $\cos \phi_1$ to $\cos \phi_2$

$$= \frac{I_C}{\omega V}$$

$$\left(\because X_C = \frac{V}{I_C} = \frac{1}{\omega C} \right)$$

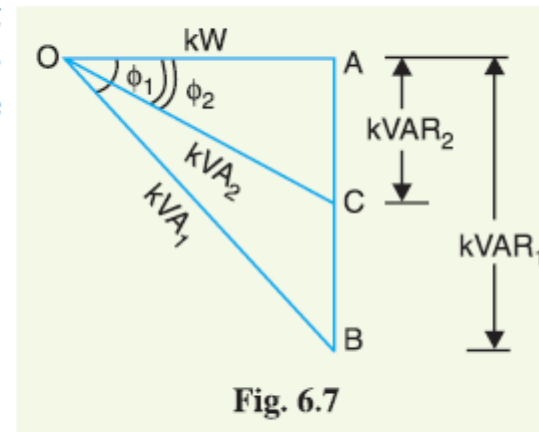
Methods for Improving Power Factor

Power triangle. The power factor correction can also be illustrated from power triangle. Thus referring to Fig. 6.7, the power triangle OAB is for the power factor $\cos \phi_1$, whereas power triangle OAC is for the improved power factor $\cos \phi_2$. It may be seen that active power (OA) does not change with power factor improvement. However, the lagging kVAR of the load is reduced by the p.f. correction equipment, thus improving the p.f. to $\cos \phi_2$.

Leading kVAR supplied by p.f. correction equipment

$$\begin{aligned} &= BC = AB - AC \\ &= \text{kVAR}_1 - \text{kVAR}_2 \\ &= OA (\tan \phi_1 - \tan \phi_2) \\ &= \text{kW} (\tan \phi_1 - \tan \phi_2) \end{aligned}$$

Knowing the leading kVAR supplied by the p.f. correction equipment, the desired results can be obtained.



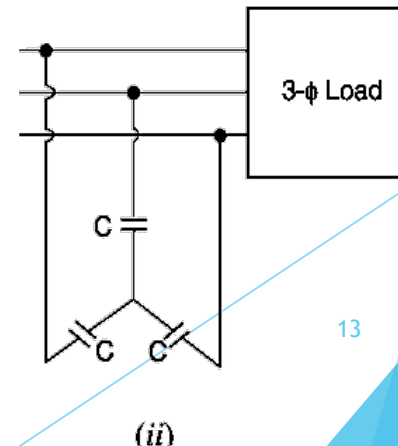
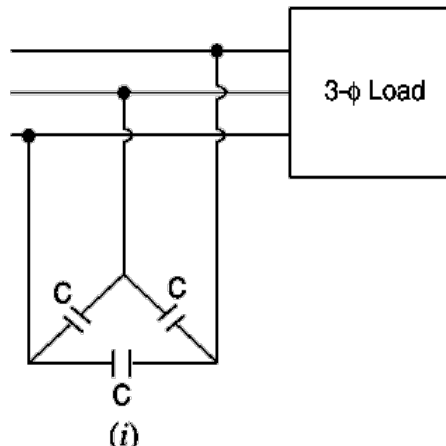
Methods for Improving Power Factor

Normally, the power factor of the whole load on a large generating station is in the region of 0.8 to 0.9. However, sometimes it is lower and in such cases it is generally desirable to take special steps to improve the power factor. This can be achieved by the following equipment:

1. Static capacitors
2. Synchronous condenser
3. Phase advancers

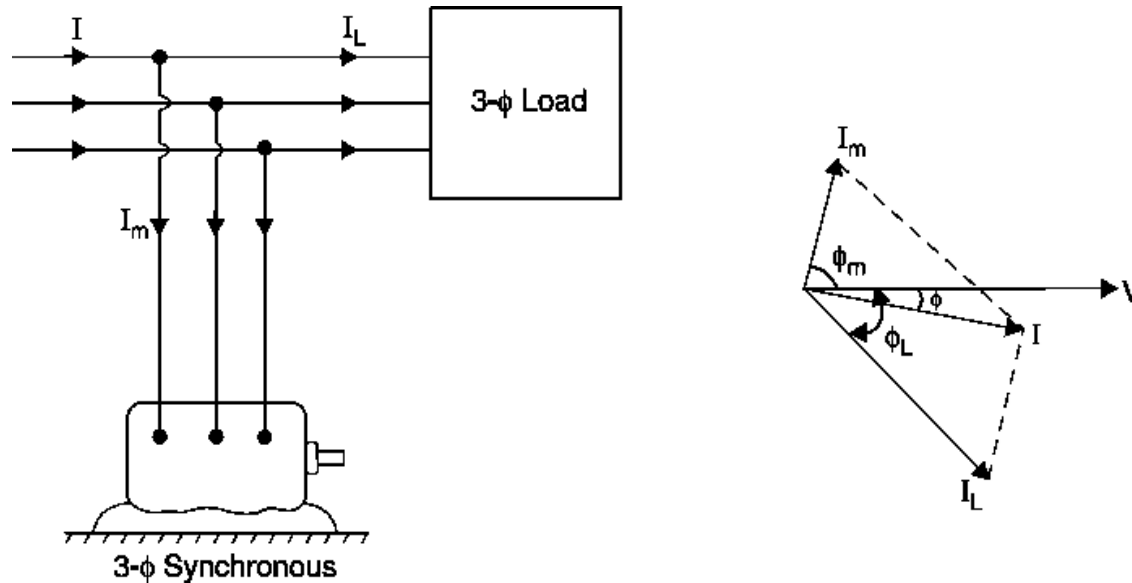
Static capacitors

The power factor can be improved by connecting capacitors in parallel with the equipment operating at lagging power factor. The capacitor (generally known as static capacitor) draws a leading current and partly or completely neutralises the lagging reactive component of load current. This raises the power factor of the load. For three-phase loads, the capacitors can be connected in delta or star as shown in Fig. Static capacitors are invariably used for power factor improvement in factories.



Methods for Improving Power Factor

Synchronous Condensers



A synchronous motor takes a leading current when over-excited and, therefore, behaves as a capacitor. An over-excited synchronous motor running on no load is known as synchronous condenser. When such a machine is connected in parallel with the supply, it takes a leading current which partly neutralises the lagging reactive component of the load. Thus the power factor is improved.

Fig shows the power factor improvement by synchronous condenser method. The 3 ϕ load takes current I_L at low lagging power factor $\cos\phi_L$. The synchronous condenser takes a current I_m which leads the voltage by an angle ϕ_m . The resultant current I is the phasor sum of I_m and I_L and lags behind the voltage by an angle ϕ . It is clear that ϕ is less than ϕ_L so that $\cos\phi$ is greater than $\cos\phi_L$. Thus the power factor is increased from $\cos\phi_L$ to $\cos\phi$. Synchronous condensers are generally used at major bulk supply substations for power factor improvement. It ensures stepless control of power factor.

Examples

Example 6.1 *An alternator is supplying a load of 300 kW at a p.f. of 0.6 lagging. If the power factor is raised to unity, how many more kilowatts can alternator supply for the same kVA loading ?*

Solution :

$$\text{kVA} = \frac{\text{kW}}{\cos \phi} = \frac{300}{0.6} = 500 \text{ kVA}$$

$$\text{kW at 0.6 p.f.} = 300 \text{ kW}$$

$$\text{kW at 1 p.f.} = 500 \times 1 = 500 \text{ kW}$$

$$\begin{aligned} \therefore \text{ Increased power supplied by the alternator} \\ = 500 - 300 = \mathbf{200 \text{ kW}} \end{aligned}$$

Note the importance of power factor improvement. When the p.f. of the alternator is unity, the 500 kVA are also 500 kW and the engine driving the alternator has to be capable of developing this power together with the losses in the alternator. But when the power factor of the load is 0.6, the power is only 300 kW. Therefore, the engine is developing only 300 kW, though the alternator is supplying its rated output of 500 kVA.

Problems

Problem 1: Let's say an industrial installation has:

- a. An active power of 100 kW ($P = 100 \text{ kW}$).
- b. An initial power factor of 0.7 ($\cos \varphi_1 = 0.7$).
- c. A desired power factor of 0.95 ($\cos \varphi_2 = 0.95$)

Determine the KVAR required.

Examples

Note the effect of connecting a $94.3 \mu\text{F}$ capacitor in parallel with the motor. The current taken from the supply is reduced from 31.7 A to 24.65 A without altering the current or power taken by the motor. This enables an economy to be effected in the size of generating plant and in the cross-sectional area of the conductors.

Example 6.2 A single phase motor connected to 400 V , 50 Hz supply takes 31.7 A at a power factor of 0.7 lagging. Calculate the capacitance required in parallel with the motor to raise the power factor to 0.9 lagging.

Solution : The circuit and phasor diagrams are shown in Figs. 6.8 and 6.9 respectively. Here motor M is taking a current I_M of 31.7 A . The current I_C taken by the capacitor must be such that when combined with I_M the resultant current I lags the voltage by an angle ϕ where $\cos \phi = 0.9$.

Referring to the phasor diagram in Fig. 6.9,

$$\text{Active component of } I_M = I_M \cos \phi_M = 31.7 \times 0.7 = 22.19 \text{ A}$$

$$\text{Active component of } I = I \cos \phi = I \times 0.9$$

These components are represented by OA in Fig. 6.9.

$$\therefore I = \frac{22.19}{0.9} = 24.65 \text{ A}$$

$$\text{Reactive component of } I_M = I_M \sin \phi_M = 31.7 \times 0.714^* = 22.6 \text{ A}$$

$$\begin{aligned} \text{Reactive component of } I &= I \sin \phi = 24.65 \sqrt{1 - (0.9)^2} \\ &= 24.65 \times 0.436 = 10.75 \text{ A} \end{aligned}$$

It is clear from Fig. 6.9 that :

$$\begin{aligned} I_C &= \text{Reactive component of } I_M - \text{Reactive component of } I \\ &= 22.6 - 10.75 = 11.85 \text{ A} \end{aligned}$$

$$\text{But } I_C = \frac{V}{X_C} = V \times 2\pi f C$$

$$\text{or } 11.85 = 400 \times 2\pi \times 50 \times C$$

$$\therefore C = 94.3 \times 10^{-6} \text{ F} = 94.3 \mu\text{F}$$

$$* \sin \phi_M = \sqrt{1 - \cos^2 \phi_M} = \sqrt{1 - (0.7)^2} = 0.714$$

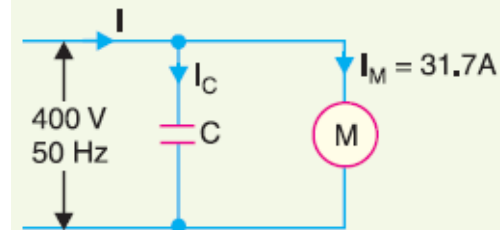


Fig. 6.8

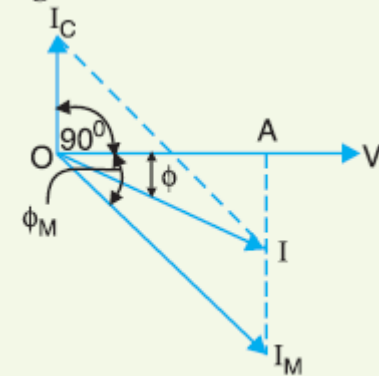


Fig. 6.9

Examples

Example 6.3 A single phase a.c. generator supplies the following loads :

- (i) Lighting load of 20 kW at unity power factor.
- (ii) Induction motor load of 100 kW at p.f. 0.707 lagging.
- (iii) Synchronous motor load of 50 kW at p.f. 0.9 leading.

Calculate the total kW and kVA delivered by the generator and the power factor at which it works.

Solution : Using the suffixes 1, 2 and 3 to indicate the different loads, we have,

$$\text{kVA}_1 = \frac{\text{kW}_1}{\cos \phi_1} = \frac{20}{1} = 20 \text{ kVA}$$

$$\text{kVA}_2 = \frac{\text{kW}_2}{\cos \phi_2} = \frac{100}{0.707} = 141.4 \text{ kVA}$$

$$\text{kVA}_3 = \frac{\text{kW}_3}{\cos \phi_3} = \frac{50}{0.9} = 55.6 \text{ kVA}$$

These loads are represented in Fig. 6.10. The three kVAs' are not in phase. In order to find the total kVA, we resolve each kVA into rectangular components – kW and kVAR as shown in Fig. 6.10. The total kW and kVAR may then be combined to obtain total kVA.

$$\text{kVAR}_1 = \text{kVA}_1 \sin \phi_1 = 20 \times 0 = 0$$

$$\text{kVAR}_2 = \text{kVA}_2 \sin \phi_2 = 141.4 \times 0.707 = -100 \text{ kVAR}$$

$$\text{kVAR}_3 = \text{kVA}_3 \sin \phi_3 = +55.6 \times 0.436 = +24.3 \text{ kVAR}$$

Note that kVAR_2 and kVAR_3 are in opposite directions ; kVAR_2 being a lagging while kVAR_3 being a leading kVAR.

$$\text{Total kW} = 20 + 100 + 50 = \mathbf{170 \text{ kW}}$$

$$\text{Total kVAR} = 0 - 100 + 24.3 = -75.7 \text{ kVAR}$$

$$\text{Total kVA} = \sqrt{(\text{kW})^2 + (\text{kVAR})^2} = \sqrt{(170)^2 + (75.7)^2} = \mathbf{186 \text{ kVA}}$$

$$\text{Power factor} = \frac{\text{Total kW}}{\text{Total kVA}} = \frac{170}{186} = \mathbf{0.914 \text{ lagging}}$$

The power factor must be lagging since the resultant kVAR is lagging.

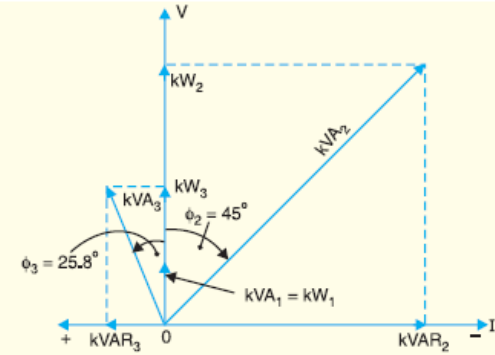


Fig. 6.10

Problems

Problem 2: A factory load consists of the following:

- a. An induction motor of 50hp with 0.8 p.f. and efficiency 0.85.
- b. A synchronous motor of 25hp with 0.9 p.f. and efficiency 0.9
- c. Lighting load of 10kW at unity p.f.

Find the input total active power, total KVA and net KVAR.

Examples

Example 6.4 A 3-phase, 5 kW induction motor has a p.f. of 0.75 lagging. A bank of capacitors is connected in delta across the supply terminals and p.f. raised to 0.9 lagging. Determine the kVAR rating of the capacitors connected in each phase.

Solution :

Original p.f., $\cos \phi_1 = 0.75$ lag ; Motor input, $P = 5$ kW

Final p.f., $\cos \phi_2 = 0.9$ lag ; Efficiency, $\eta = 100$ % (assumed)

$$\phi_1 = \cos^{-1}(0.75) = 41.41^\circ ; \tan \phi_1 = \tan 41.41^\circ = 0.8819$$

$$\phi_2 = \cos^{-1}(0.9) = 25.84^\circ ; \tan \phi_2 = \tan 25.84^\circ = 0.4843$$

Leading kVAR taken by the condenser bank

$$= P (\tan \phi_1 - \tan \phi_2)$$

$$= 5 (0.8819 - 0.4843) = 1.99 \text{ kVAR}$$

$\therefore$ Rating of capacitors connected in each phase

$$= 1.99/3 = \mathbf{0.663 \text{ kVAR}}$$

Please check examples 6.1-6.6,6.9,6.11, tutorial problems- 1-3, 5,6, Mehta.

Reference:

Principles of Power System by V.K. Mehta and Rohit Mehta



QUESTIONS AND DISCUSSIONS

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